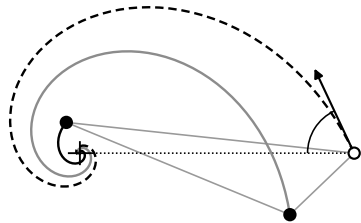


(a) $\mu = 1/2, \quad P = 1.0647\dots$

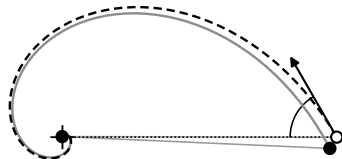


— $\Gamma_1 = 1$ --- $\Gamma_3 = -1/3$
 — $\Gamma_2 = 1/2$ r_0 (third vortex)

$$\arctan 2P = 64.8^\circ > 60^\circ$$

$$\text{path length } r_0 \sqrt{1 + 4P^2} = 2.35 r_0 > 2r_0$$

(b) $\mu = 0.05, \quad P = 0.8691\dots$



— $\Gamma_1 = 1$ --- $\Gamma_3 = -1/21$
 — $\Gamma_2 = 0.05$ r_0 (third vortex)

$$\arctan 2P = 60.1^\circ > 60^\circ$$

$$\text{path length } r_0 \sqrt{1 + 4P^2} = 2.005 r_0 > 2r_0$$